Description

Image





Caption

1. Butyl rubber is one of the most important materials for inner tubes. © ANSYS, Inc. 2. Butyl Rubber (Viton®) Gloves © Ansell

The material

Butyl Rubbers (IIR) are synthetics that resemble natural rubber (NR) in properties. They have good resistance to abrasion, tearing and flexing, with exceptionally low gas permeability and useful properties up to 150 C. They have low dielectric constant and loss, making them attractive for electrical applications.

Compositional summary

(CH2-C(CH3)-CH-(CH2)2-C(CH3)2)n

General properties

Density		910	-	950	kg/m^3
Price	*	1.94	-	2.05	GBP/kg
Date first used		1937			

Mechanical properties

Young's modulus		7e-4	-	0.0015	GPa
Shear modulus		2.34e-4	-	5e-4	GPa
Bulk modulus	*	1.5	-	2	GPa
Poisson's ratio		0.499	-	0.5	
Yield strength (elastic limit)		2.4	-	10	MPa
Tensile strength		2.4	-	10	MPa
Compressive strength	*	2.88	-	12	MPa
Elongation		480	-	950	% strain
Hardness - Vickers	*	2	-	4	HV

Fatigue strength at 10^7 cycles	*	0.96	-	4	MPa
Fracture toughness		0.037	-	0.111	MPa.m^0.5
Mechanical loss coefficient (tan delta)		0.5	-	0.9	

Thermal properties

Glass temperature		-73.2	-	-63.2	°C
Maximum service temperature		96.9	-	120	°C
Minimum service temperature		-51.2	-	-25.2	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.08	-	0.13	W/m.°C
Specific heat capacity		1.85e3	-	1.95e3	J/kg.°C
Thermal expansion coefficient		170	-	200	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		1e22	-	1e23	µohm.cm
Dielectric constant (relative permittivity)		2.1	-	2.6	
Dissipation factor (dielectric loss tangent)		0.003	-	0.0054	
Dielectric strength (dielectric breakdown)		16	-	24	MV/m

Optical properties

Transparency		Translucent			
Refractive index		1.51	-	1.52	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		4	-	5	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		1			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Acceptable			
Soils, alkaline (clay)		Limited use			
Wine		Excellent			

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Limited use
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Excellent
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Excellent

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Unacceptable
Benzene	Unacceptable
Carbon tetrachloride	Unacceptable
Chloroform	Unacceptable
Crude oil	Unacceptable
Diesel oil	Unacceptable
Lubricating oil	Unacceptable
Paraffin oil (kerosene)	Unacceptable
Petrol (gasoline)	Unacceptable
Silicone fluids	Excellent
Toluene	Unacceptable
Turpentine	Unacceptable
Vegetable oils (general)	Limited use
White spirit	Unacceptable

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	

	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Poor

Durability: flammability

Flammability	Highly flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	1.03e7	tonne/yr
Reserves, principal component	2.9e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 4.23	-	4.67	kg/kg
Embodied energy (virgin grade production)	* 90.3	-	99.5	MJ/kg
Water usage	* 63.8	-	191	l/kg

Material processing: energy

Polymer molding (injection, blow), embodied energy	* 16.1	-	17.7	MJ/kg
Machining, grinding, embodied energy	* 15.5	-	20.2	MJ/kg

Material processing: climate change

Polymer molding (injection, blow), climate change (CO2-eq)	* 1.05	-	1.16	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.02	-	1.32	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Combust for energy recovery	✓			
Heat of combustion (net)	* 43.2	-	45.3	MJ/kg
Combustion CO2	* 3.11	-	3.27	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Butyl elastomers are thermosets, and thus cannot be recycled. Their disposal creates an environmental problem.

Recycle mark



Supporting information

Typical uses

Inner tubes, seals, belts, anti-vibration mounts, electrical insulation, tubing, brake pads, rubber lining pipes and pumps.

Links

ProcessUniverse

Producers

Reference